

Orbital Rotation Gradient and Hessian for Exponential Parameterization (RDMFT-style, expressed via the Generalized Fock Matrix)

Conventions

One-particle and two-particle reduced density matrices:

$$D_{ab} = \langle 0 | a_a^\dagger a_b | 0 \rangle, \quad (1)$$

$$\Gamma_{pqrs} = \frac{1}{2} \langle 0 | a_p^\dagger a_q^\dagger a_s a_r | 0 \rangle. \quad (2)$$

Two-electron integrals in physicist's notation, $\langle ik|jl \rangle$.

Hamiltonian:

$$\hat{H} = \hat{H}_1 + \hat{H}_2, \quad \hat{H}_1 = \sum_{pq} h_{pq} a_p^\dagger a_q, \quad \hat{H}_2 = \frac{1}{2} \sum_{uvwx} \langle uv|wx \rangle a_u^\dagger a_v^\dagger a_x a_w. \quad (3)$$

Orbital rotation generator (anti-Hermitian): $\hat{\kappa} = \sum_{pq} \kappa_{pq} a_p^\dagger a_q$, with $\kappa_{qp} = -\kappa_{pq}^*$.

Generalized Fock matrix

$$F_{qp} = \sum_r D_{rq} h_{rp} + 2 \sum_{rst} \Gamma_{srtq} \langle sr|tp \rangle. \quad (4)$$

Gradient

$$g_{pq} = F_{qp} - F_{pq}^*. \quad (5)$$

Hessian

Energy expansion:

$$E(\boldsymbol{\kappa}) \approx E(\mathbf{0}) + \sum_{pq} \kappa_{pq} g_{pq} + \frac{1}{2} \sum_{pq,rs} \kappa_{pq} \kappa_{rs} G_{pq,rs} + \dots, \quad G_{pq,rs} = \langle 0 | [[\hat{H}, a_p^\dagger a_q], a_r^\dagger a_s] | 0 \rangle. \quad (6)$$

One-electron contribution

$$G_{pq,rs}^{(1)} = \delta_{qr} \sum_u h_{up} D_{us} - h_{sp} D_{rq} - h_{qr} D_{ps} + \delta_{sp} \sum_v h_{qv} D_{rv}. \quad (7)$$

Two-electron contribution

$$\begin{aligned}
G_{pq,rs}^{(2)} = & + 2 \delta_{rq} \sum_{uvt} \langle uv|pt \rangle \Gamma_{uvst} - 2 \sum_{uv} \langle uv|pr \rangle \Gamma_{uvsq} \\
& + 2 \sum_{ut} \langle us|pt \rangle \Gamma_{ruqt} - 2 \sum_{ut} \langle su|pt \rangle \Gamma_{ruqt} \\
& + 2 \sum_{ut} \langle uq|rt \rangle \Gamma_{pust} - 2 \sum_{uw} \langle uq|wr \rangle \Gamma_{pusw} \\
& + 2 \sum_{wt} \langle sq|wt \rangle \Gamma_{rpwt} - 2 \delta_{sp} \sum_{uwt} \langle uq|wt \rangle \Gamma_{ruwt}.
\end{aligned} \tag{8}$$

Full Hessian, Fock-matrix form

The δ_{qr} and δ_{sp} terms of $G^{(1)}$ and $G^{(2)}$ combine exactly into F_{sp} and F_{rq}^* , leaving six two-electron “exchange-type” terms that do not reduce to F :

$$\begin{aligned}
G_{pq,rs} = & \delta_{qr} F_{sp} + \delta_{sp} F_{rq}^* - h_{sp} D_{rq} - h_{qr} D_{ps} \\
& - 2 \sum_{uv} \langle uv|pr \rangle \Gamma_{uvsq} + 2 \sum_{ut} \langle us|pt \rangle \Gamma_{ruqt} - 2 \sum_{ut} \langle su|pt \rangle \Gamma_{ruqt} \\
& + 2 \sum_{ut} \langle uq|rt \rangle \Gamma_{pust} - 2 \sum_{uw} \langle uq|wr \rangle \Gamma_{pusw} + 2 \sum_{wt} \langle sq|wt \rangle \Gamma_{rpwt}
\end{aligned} \tag{9}$$

Real/imaginary directional blocks

For a complex (relativistic, spin-orbit-coupled) spinor basis, κ_{pq} for an independent pair $p > q$ is a free complex number, $\kappa_{pq} = t_{pq} + iy_{pq}$ ($\kappa_{qp} = -t_{pq} + iy_{pq}$, enforcing $\kappa_{qp} = -\kappa_{pq}^*$). The full real Hessian over these $2 \times (\text{number of pairs})$ real parameters therefore has three independent blocks between any two distinct pairs (p, q) and (r, s) , all obtainable from the SAME four raw terms $G_{pq,rs}$, $G_{pq,sr}$, $G_{qp,rs}$, $G_{qp,sr}$ by substituting $\kappa_{pq} = t + iy$, $\kappa_{qp} = -t + iy$, $\kappa_{rs} = t' + iy'$, $\kappa_{sr} = -t' + iy'$ into $\frac{1}{2} \sum \kappa_{pq} \kappa_{rs} G_{pq,rs}$ and reading off the coefficient of each real product:

$$\text{Hess}_{pq,rs} = \frac{\partial^2 E}{\partial t_{pq} \partial t_{rs}} = G_{pq,rs} - G_{pq,sr} - G_{qp,rs} + G_{qp,sr}, \tag{10}$$

$$\text{Hess}_{pq,rs}^{yy} = \frac{\partial^2 E}{\partial y_{pq} \partial y_{rs}} = -(G_{pq,rs} + G_{pq,sr} + G_{qp,rs} + G_{qp,sr}), \tag{11}$$

$$\text{Hess}_{pq,rs}^{ty} = \frac{\partial^2 E}{\partial t_{pq} \partial y_{rs}} = i(G_{rs,pq} + G_{rs,qp} - G_{sr,pq} - G_{sr,qp}). \tag{12}$$

The first two are the $(+, -, -, +)$ and $(-, -, -, -)$ sign patterns over the four raw terms with (p, q) in the flipping (first) slot; the third is the same idea but with the *other* pair, (r, s) , in the flipping slot – not (p, q) , which is easy to get backwards by analogy with the first two (an earlier draft of this derivation did exactly that, and was only caught by the finite-difference check below). The fourth possible direction, $\text{Hess}_{pq,rs}^{yt} = \partial^2 E / \partial y_{pq} \partial t_{rs}$, is not a new formula: by Schwarz’s theorem, $\text{Hess}_{pq,rs}^{yt} = \text{Hess}_{rs,pq}^{ty}$.

All three are real for physical (Hermitian h , $\langle \cdot | \cdot \rangle$, D , Γ) input by the same argument: $K_{pq} = a_p^\dagger a_q - a_q^\dagger a_p$ and $K'_{rs} = i(a_r^\dagger a_s + a_s^\dagger a_r)$ are both anti-Hermitian, so $[[\hat{H}, K_{pq}], K'_{rs}]$ is Hermitian and its expectation value is real, regardless of which of the three sign patterns above it corresponds to.

Status note (updated). $G_{pq,rs}^{(1)}$, the gradient g_{pq} , $\text{Hess}_{pq,rs}$, and $\text{Hess}_{pq,rs}^{yy}$ have all been independently verified: by more than one derivation route, by an explicit small-Fock-space numerical check against a general (non-idempotent, non-Wick-derivable) random 1-/2-RDM, and by finite-difference checks against converged NON_REL/C4_DHF SCF data. $\text{Hess}_{pq,rs}^{ty}$ (and hence $\text{Hess}_{pq,rs}^{yt}$ via Schwarz) has since also been derived and confirmed the same way – a genuine mixed-direction (real step on one pair, imaginary step on the other) 2D finite difference against water/STO-3G's converged AND a deliberately non-converged (1 SCF iteration) C4_DHF density, matching to $O(\text{step}^2)$ in both cases (implementation: `Hessian_opt/HartreeExchangeHessian.h`'s `hartreeExchangeHessianElementMixed`, the Hartree/exchange-ansatz "cheap" path only – the fully general (`GeneralizedHessian.h`) counterpart has not been added). The only remaining unimplemented piece for a full $2N(N-1)$ -real-parameter orbital-rotation Hessian is the general (non-Hartree/exchange-ansatz) 2-RDM version of the mixed block, and assembling all three blocks into one joint matrix for an actual Newton-Raphson/NEO step (currently only $\text{Hess}_{pq,rs}$ alone is ever assembled into a diagonalizable matrix, for the minimum/saddle-point diagnostic in `main.cpp`'s `HESSIAN_4C`).